ELECTRIC VEHICLES, BATTERIES AND ALTERNATIVE FUELS

POWERTRAIN SOURCE-OF-ELECTRICITY MAP → CELL TO PACK SYSTEMS → LITHIUM-ION DISCHARGE RAIL → BATTERY CHEMISTRY DECISION MATRIX → ENERGY AND POWER AXES → AC AND DC CHARGING

Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

◆ PRIMARY CORE FLOW ◆ SUBORDINATE EXTRA ◆ APPROVAL / REVIEW

CORE BEFORE EXTRA • prior artefacts unchanged • exact same master drives poster and tiles

Approval: FALSE • Pending user review • active generation g1 • explicit approval of this exact generation is required • all prior artifacts unchanged

00

STAGE 00

POWERTRAIN SOURCE-OF-ELECTRICITY MAP

POWERTRAIN SOURCE-OF-ELECTRICITY MAP · BATTERY STORES ALL TRACTION ELECTRICITY · ENGINE + REGENERATIVE BRAKING CHARGE NON-PLUG-IN BATTERY · EXTERNAL PLUG + ENGINE BACKUP · HYDROGEN FUEL CELL GENERATES ONBOARD ELECTRICITY · WHERE DOES TRACTION ELECTRICITY ORIGINATE?

BEV → battery stores all traction electricity

HEV → engine + regenerative braking charge non-plug-in battery → **PHEV → external plug + engine backup** → **FCEV → hydrogen fuel cell generates onboard electricity**

CONCEPT / ARCHITECTURE

- BEV → battery stores all traction electricity
- HEV → engine + regenerative braking charge non-plug-in battery

MECHANISM / APPLICATION

- PHEV → external plug + engine backup
- FCEV → hydrogen fuel cell generates onboard electricity

READINESS / STATUS LIMIT

- TEST → where does traction electricity originate?
- BEV → battery stores all traction electricity

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: FCEV → hydrogen fuel cell generates onboard electricity.

01

STAGE 01

CELL TO PACK SYSTEMS HIERARCHY

CELL TO PACK SYSTEMS HIERARCHY · ELECTROCHEMICAL UNIT · GROUPED CELLS + MECHANICAL INTEGRATION · MODULES/CELLS + ENCLOSURE + BUSBARS + SENSING · BMS + CONTACTORS + PROTECTION · COOLING/HEATING + PROPAGATION BARRIERS

CELL → electrochemical unit

MODULE → grouped cells + mechanical integration → **PACK → modules/cells + enclosure + busbars + sensing** → **CONTROL → BMS + contactors + protection**

CONCEPT / ARCHITECTURE

- CELL → electrochemical unit
- MODULE → grouped cells + mechanical integration

MECHANISM / APPLICATION

- PACK → modules/cells + enclosure + busbars + sensing
- CONTROL → BMS + contactors + protection

READINESS / STATUS LIMIT

- THERMAL → cooling/heating + propagation barriers
- CELL → electrochemical unit

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: PACK → modules/cells + enclosure + busbars + sensing.

02

STAGE 02

LITHIUM-ION DISCHARGE RAIL

LITHIUM-ION DISCHARGE RAIL · RELEASES Li+ DURING DISCHARGE · CARRIES IONS · BLOCKS DIRECT ELECTRONIC CONTACT · RECEIVES Li+ DURING DISCHARGE · EXTERNAL CIRCUIT · ELECTRONS POWER MOTOR

ANODE → releases Li+ during discharge → **ELECTROLYTE → carries ions** → **SEPARATOR → blocks direct electronic contact** → **CATHODE → receives Li+ during discharge** → **EXTERNAL CIRCUIT → electrons power motor**

STARTING CONDITION / INPUT

- ANODE → releases Li+ during discharge
- ELECTROLYTE → carries ions

SCIENTIFIC OR ENGINEERING MECHANISM

- SEPARATOR → blocks direct electronic contact
- CATHODE → receives Li+ during discharge

OUTPUT / FAILURE BOUNDARY

- EXTERNAL CIRCUIT → electrons power motor
- ANODE → releases Li+ during discharge

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CATHODE → receives Li+ during discharge; EXTERNAL CIRCUIT → electrons power motor.

03

STAGE 03

BATTERY CHEMISTRY DECISION MATRIX

BATTERY CHEMISTRY DECISION MATRIX · NICKEL-COBALT EXPOSURE; HIGHER-DENSITY DESIGNS · NO NICKEL OR COBALT; DIFFERENT DENSITY/SAFETY TRADE-OFF · CHANGES LITHIUM DEPENDENCE · SOLID ELECTROLYTE; PRE-COMMERCIAL BOUNDARY · CHEMISTRY DECIDES MINERAL BILL AND ENGINEERING TRADE-OFF

SYSTEM / DEVICE / INSTITUTION	DECISIVE TECHNICAL DISTINCTION	STATUS / UNIT / EVIDENCE LIMIT
NMC / NCA → nickel-cobalt exposure; higher-density designs	SODIUM-ION → changes lithium dependence	RULE → chemistry decides mineral bill and engineering trade-off
LFP → no nickel or cobalt; different density/safety trade-off	SOLID-STATE → solid electrolyte; pre-commercial boundary	NMC / NCA → nickel-cobalt exposure; higher-density designs

SYSTEM / DEVICE / INSTITUTION

- NMC / NCA → nickel-cobalt exposure; higher-density designs
- LFP → no nickel or cobalt; different density/safety trade-off

DECISIVE TECHNICAL DISTINCTION

- SODIUM-ION → changes lithium dependence
- SOLID-STATE → solid electrolyte; pre-commercial boundary

STATUS / UNIT / EVIDENCE LIMIT

- RULE → chemistry decides mineral bill and engineering trade-off
- NMC / NCA → nickel-cobalt exposure; higher-density designs

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE → chemistry decides mineral bill and engineering trade-off.

04

STAGE 04

ENERGY AND POWER AXES

ENERGY AND POWER AXES · ENERGY DENSITY · HOW MUCH ENERGY PER MASS/VOLUME · POWER DENSITY · HOW QUICKLY ENERGY IS DELIVERED/ACCEPTED · ENERGY AXIS · FAST CHARGE · POWER AXIS

ENERGY DENSITY → how much energy per mass/volume

POWER DENSITY → how quickly energy is delivered/accepted → **RANGE / WEIGHT → energy axis** → **ACCELERATION / REGEN / FAST CHARGE → power axis**

CONCEPT / ARCHITECTURE

- ENERGY DENSITY → how much energy per mass/volume
- POWER DENSITY → how quickly energy is delivered/accepted

MECHANISM / APPLICATION

- RANGE / WEIGHT → energy axis
- ACCELERATION / REGEN / FAST CHARGE → power axis

READINESS / STATUS LIMIT

- TRAP → one metric cannot certify whole-pack performance
- ENERGY DENSITY → how much energy per mass/volume

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: TRAP → one metric cannot certify whole-pack performance.

05

STAGE 05

AC AND DC CHARGING CONVERSION MAP

AC AND DC CHARGING CONVERSION MAP · GRID AC · VEHICLE ONBOARD CHARGER · REGULATED BATTERY DC · OFFBOARD FAST CHARGER · POWER AND USE CONTEXT · PHYSICAL/ELECTRICAL INTERFACE · COMMUNICATION AND CONTROL

GRID AC → vehicle onboard charger → regulated battery DC

GRID AC → offboard fast charger → regulated battery DC → **LEVEL → power and use context** → **CONNECTOR → physical/electrical interface**

STARTING CONDITION / INPUT

- GRID AC → vehicle onboard charger → regulated battery DC
- GRID AC → offboard fast charger → regulated battery DC

SCIENTIFIC OR ENGINEERING MECHANISM

- LEVEL → power and use context
- CONNECTOR → physical/electrical interface

OUTPUT / FAILURE BOUNDARY

- PROTOCOL → communication and control
- GRID AC → vehicle onboard charger → regulated battery DC

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: GRID AC → offboard fast charger → regulated battery DC.

06

STAGE 06

CHARGING GOVERNANCE INSTITUTION MAP

CHARGING GOVERNANCE INSTITUTION MAP · MINISTRY OF POWER · CHARGING GUIDELINES · CONNECTOR AND SAFETY STANDARDS · ELECTRICITY-SYSTEM AND ELECTRICAL SAFETY REGULATION · VEHICLE APPROVAL AND SAFETY · EFFICIENCY ROLE

MINISTRY OF POWER → charging guidelines

BIS → connector and safety standards → **CEA → electricity-system and electrical safety regulation** → **MORTH / CMVR → vehicle approval and safety**

CONCEPT / ARCHITECTURE

- MINISTRY OF POWER → charging guidelines
- BIS → connector and safety standards

MECHANISM / APPLICATION

- CEA → electricity-system and electrical safety regulation
- MORTH / CMVR → vehicle approval and safety

READINESS / STATUS LIMIT

- BEE → efficiency role
- MINISTRY OF POWER → charging guidelines

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: MORTH / CMVR → vehicle approval and safety.

07

STAGE 07

BMS AND THERMAL-CONTROL LOOP

BMS AND THERMAL-CONTROL LOOP · OPERATING STATE AND IMBALANCE · KEEP CELLS IN SUITABLE BAND · SAFER OPERATION AND SLOWER DEGRADATION, NOT IMMUNITY

CONCEPT / ARCHITECTURE

- SENSE → voltage
- ESTIMATE → operating state and imbalance

MECHANISM / APPLICATION

- ACT → balance
- COOL / HEAT → keep cells in suitable band

READINESS / STATUS LIMIT

- FEEDBACK → safer operation and slower degradation, not immunity
- SENSE → voltage

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: FEEDBACK → safer operation and slower degradation, not immunity.

08

STAGE 08

DEGRADATION AND SAFETY FAULT TREE

DEGRADATION AND SAFETY FAULT TREE · HIGH SOC · DEEP CYCLING · HIGH RATE · AGEING PRESSURE · LOCAL HEAT GENERATION · SELF-HEATING HEAT REMOVAL · THERMAL RUNAWAY

HEAT / HIGH SOC / DEEP CYCLING / HIGH RATE → ageing pressure

DEFECT / ABUSE / SHORT → local heat generation → **SELF-HEATING HEAT REMOVAL → thermal runaway** → **CELL EVENT → possible module/pack propagation**

CONCEPT / ARCHITECTURE

- HEAT / HIGH SOC / DEEP CYCLING / HIGH RATE → ageing pressure
- DEFECT / ABUSE / SHORT → local heat generation

MECHANISM / APPLICATION

- SELF-HEATING HEAT REMOVAL → thermal runaway
- CELL EVENT → possible module/pack propagation

READINESS / STATUS LIMIT

- RESPONSE → prevent + detect + contain + emergency action
- HEAT / HIGH SOC / DEEP CYCLING / HIGH RATE → ageing pressure

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RESPONSE → prevent + detect + contain + emergency action.

09

STAGE 09

HYDROGEN MOBILITY CHAIN

HYDROGEN MOBILITY CHAIN · PRODUCTION ROUTE · GREY LABEL · UPSTREAM BURDEN · PEM FUEL CELL · ELECTROCHEMICAL ELECTRICITY · ELECTRIC MOTOR · DIRECT EXHAUST

PRODUCTION ROUTE → green / blue / grey label → **COMPRESSION / TRANSPORT / STORAGE → upstream burden** → **PEM FUEL CELL → electrochemical electricity** → **ELECTRIC MOTOR → traction** → **DIRECT EXHAUST → water + heat; lifecycle remains wider**

STARTING CONDITION / INPUT

- PRODUCTION ROUTE → green / blue / grey label
- COMPRESSION / TRANSPORT / STORAGE → upstream burden

SCIENTIFIC OR ENGINEERING MECHANISM

- PEM FUEL CELL → electrochemical electricity
- ELECTRIC MOTOR → traction

OUTPUT / FAILURE BOUNDARY

- DIRECT EXHAUST → water + heat; lifecycle remains wider
- PRODUCTION ROUTE → green / blue / grey label

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: DIRECT EXHAUST → water + heat; lifecycle remains wider.

10

STAGE 10

ALTERNATIVE-FUEL PORTFOLIO

ALTERNATIVE-FUEL PORTFOLIO · PETROL-BLEND ENGINE PATHWAY · DIESEL-BLEND PATHWAY · PURIFIED BIOGAS PATHWAY · GASEOUS COMBUSTION FUELS · HYDROGEN FUEL CELL · ELECTROCHEMICAL ELECTRIC-DRIVE PATHWAY

ETHANOL / FLEX-FUEL → petrol-blend engine pathway

BIODIESEL → diesel-blend pathway → **CBG / SATAT → purified biogas pathway** → **CNG / LNG / HCNG → gaseous combustion fuels**

CONCEPT / ARCHITECTURE

- ETHANOL / FLEX-FUEL → petrol-blend engine pathway
- BIODIESEL → diesel-blend pathway

MECHANISM / APPLICATION

- CBG / SATAT → purified biogas pathway
- CNG / LNG / HCNG → gaseous combustion fuels

READINESS / STATUS LIMIT

- HYDROGEN FUEL CELL → electrochemical electric-drive pathway
- ETHANOL / FLEX-FUEL → petrol-blend engine pathway

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CNG / LNG / HCNG → gaseous combustion fuels.

11

STAGE 11

INDIA TRANSITION AND EVIDENCE LADDER

INDIA TRANSITION AND EVIDENCE LADDER · END OF LIFE · PM E-DRIVE · DEMAND/ECOSYSTEM INSTRUMENT · PM-EBUS SEWA · NGHM PILOTS · DISTINCT ROUTES

TAILPIPE → GRID → MANUFACTURE → END OF LIFE → **MINERALS → CHEMISTRY → RECYCLING / EPR** → **PM E-DRIVE → demand/ecosystem instrument** → **PLI-AUTO / PLI-ACC / PM-eBUS SEWA / NGHM PILOTS → distinct routes** → **APPROVAL → NOTIFICATION → AWARD → COMMISSIONING → DEPLOYMENT → OUTCOME**

STARTING CONDITION / INPUT

- TAILPIPE → GRID → MANUFACTURE → END OF LIFE
- MINERALS → CHEMISTRY → RECYCLING / EPR

SCIENTIFIC OR ENGINEERING MECHANISM

- PM E-DRIVE → demand/ecosystem instrument
- PLI-AUTO / PLI-ACC / PM-eBUS SEWA / NGHM PILOTS → distinct routes

OUTPUT / FAILURE BOUNDARY

- APPROVAL → NOTIFICATION → AWARD → COMMISSIONING → DEPLOYMENT → OUTCOME
- TAILPIPE → GRID → MANUFACTURE → END OF LIFE

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: APPROVAL → NOTIFICATION → AWARD → COMMISSIONING → DEPLOYMENT → OUTCOME.

E

EXTRA

SUBORDINATE ENRICHMENT — ONLY AFTER THE COMPLETE CORE

OPTIONAL ADVANCED ENRICHMENT · THE QUESTION IS NOT "EV VERSUS NO EV"; IT · BEVS USUALLY MAKE STRONGEST SENSE WHERE DAILY DRIVING RANGE · HYDROGEN MOBILITY IS ANALYTICALLY STRONGER FOR HARD-TO-ELECTRIFY SEGMENTS THAN · ROUTINE SMALL PERSONAL TRANSPORT, THOUGH COSTS AND INFRASTRUCTURE REMAIN · ETHANOL/CNG/LNG MATTER BECAUSE THE MOBILITY TRANSITION IS BROADER THAN · MINISTRY OF HEAVY INDUSTRIES · PM E-DRIVE AND ACC BATTERY INDUSTRIAL-POLICY ANCHOR.

OPTIONAL SCIENTIFIC / ENGINEERING DEPTH

- CAUTION: The question is not "EV versus no EV"; it is which powertrain fits which transport segment .
- CAUTION: BEVs usually make strongest sense where daily driving range is moderate and charging access can be organized.
- CAUTION: Hydrogen mobility is analytically stronger for hard-to-electrify segments than for
- routine small personal transport, though costs and infrastructure remain difficult.

READINESS, UNIT OR STATUS LIMIT

- CAUTION: Ethanol/CNG/LNG matter because the mobility transition is broader than battery electrification alone.
- Ministry of Heavy Industries: PM E-DRIVE and ACC battery industrial-policy anchor.
- PM E-DRIVE portal: official implementation interface and public scheme reference.
- ACC battery PLI framework: manufacturing-capacity building, not merely demand subsidy.

COMPARATIVE TECHNOLOGY / GOVERNANCE NUANCE

- Ministry of Power / BEE / BIS / CEA: charging guidelines, standards and power-system interface.
- MNRE: green-hydrogen mission and transport pilot guidelines.
- MoPNG: ethanol-blending and alternative-fuel ecosystem.
- EVs can reduce urban tailpipe emissions and fuel-import vulnerability in selected segments.

OPTIONAL STATUS — This optional depth is unnecessary for a competent core answer; use it only for added nuance after the complete core has been written.

END OF MASTER CANVAS • poster embeds this exact master • tiled pages are overlapping crops of the same pixels